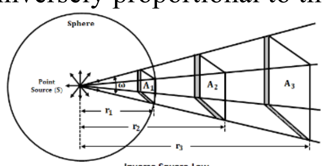
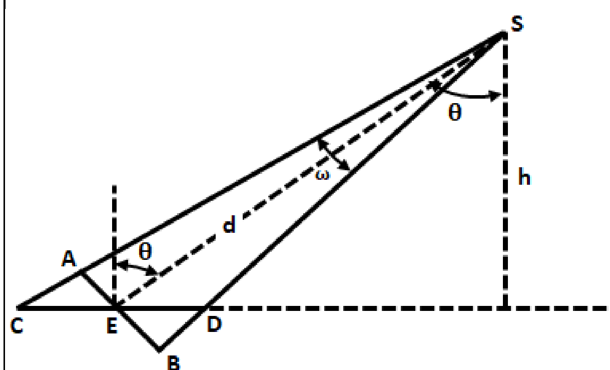
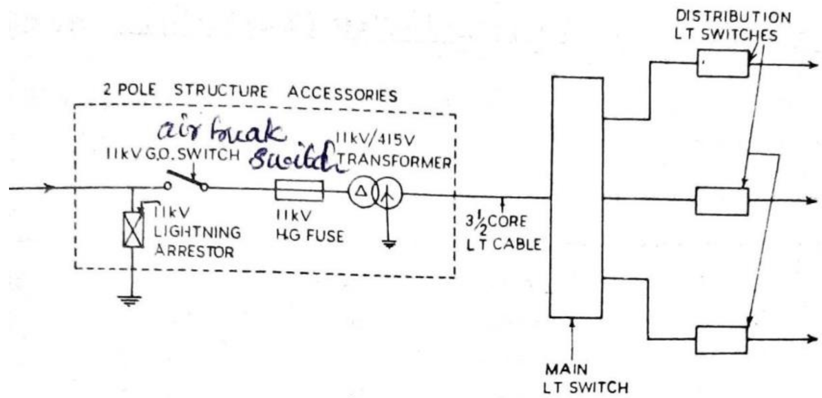
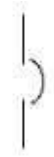
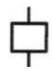
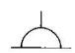
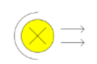
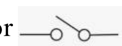
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023(2019 Scheme)			
Course Code: EET402			
Course Name: ELECTRICAL SYSTEM DESIGN AND ESTIMATION			
Max. Marks: 100			Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks.	Marks
1		1. Standard good practices for selection of part of power systems 2. Recommendations concerning safety and related matter in the wiring of electrical installations of buildings or industrial structures, promoting compatibility between such recommendations and those concerning the equipment installed 3. General safety procedures and practices in electrical work Any 3 points	(3)
2		For single phase AC: 240V, 50 Hz, 2 wire For three phase AC: 415V, 50 Hz, 4 wire Low voltage- Not exceeding 250V Voltage limits for AC systems The supply authorities are required to maintain the voltages on the system under normal condition within the tolerances specified below: • 6% in case of low and medium voltage installations • Under Indian Electricity Rules, the voltage fluctuation may not vary by more than 5% above or below the declared nominal voltage. ie. 228V to 252V for nominal voltage of 240V & 394.25V to 435.75V for nominal voltage of 415V • Frequency must be within ±1% of the declared frequency of 50Hz	(3)
3		(i) Luminous intensity (I): The power or strength of the source of light is known as luminous intensity (I) and is measured in Candela. (ii) Luminous Flux (L) : From the light source, energy is radiated in the form of	(3)

		light waves. This flow of light from the source is known as Luminous Flux (L) and unit is lumen (iii) The Steradian: This is like a three-dimensional radian, sometimes called the unit solid angle. The steradian is the solid angle subtended at the centre of a sphere by surface areas equal to r^2.	
4		Inverse Square Law: This law states that the intensity of illumination is inversely proportional to the square of the distance from the source of light.  Inverse Square Law Lambert's Cosine Law: Lambert's law describes how the illumination of a surface varies with the angle of incidence of light.  Lambert's Cosine Law	(3)
5		A residual current circuit breaker (RCCB) is an electrical safety device that detects and interrupts an electrical circuit when there is a leakage current to the ground. It protects people and equipment from electric shocks, fires, and other hazards caused by faulty wiring, insulation failure, or accidental contact with live parts. RCCB consist of a small current transformer surrounding live and neutral wire. The sensing coil on the current transformer is connected to a trip coil of a circuit breaker. Under a normal condition the current in a line conductor(IL) is same as the current in Neutral conductor (IN). Hence IL- IN is Zero, therefore two fluxes produced by IL and IN cancel each other and sensing coil does not sense any imbalance. If there is fault the fault current IF flows through the earth conductor hence there is difference between the current IL and IN. The difference IL –IN is called residual Current .The fluxes produced by IL and IN are no longer same under a fault condition, producing flux in the core. Due to the residual flux EMF get induced in the sensing coil, which circulates the current through the tripping coil of the circuit breaker.	(3)
6		No. Of light sub-circuits = 5 No. Of power sub-circuits = 3 Total number of sub-circuits = 8	(3)

7			(3)
8		Cable Sizing: Use the motor's full load current (FLC) to determine the cable size. The National Electrical Code (NEC) or local electrical codes often provide guidelines for cable sizing. Conductor Material: Choose between copper and aluminum conductors based on cost, conductivity, and the environment. Copper is a better conductor but more expensive. Insulation Type: Select the appropriate insulation type based on the environmental conditions. For example, use moisture-resistant insulation if the installation is in a damp environment. Voltage Drop: Consider the allowable voltage drop. Excessive voltage drop can affect the motor's performance. Calculate and ensure the voltage drop is within acceptable limits.	(3)
9		Continuous power rating-engine can supply rated power for an unlimited time Prime power rating: engine can supply a base load for an unlimited time, and 100% rated power for a limited time. Typical values are a base load of 70% of the rated power, 100% rated power during 500 hours per year Standby power rating: Maximum power that the engine can deliver and is limited in time, less than 500 hours per year	(3)
10		1. improve the power factor and efficiency of induction motor loads 2. Can avoid the penalties of supply authority 3. Manual switching of capacitor banks are very difficult and not practicable.	(3)

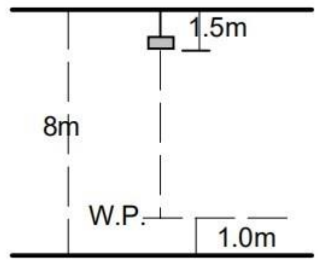
PART B			
Answer one full question from each module, each carries 14 marks.			
		Module I	
11	a)	(i)  or  (ii)  (iii)  (iv)  (v)  or  (vi)  or 	(6)
	b)	(i) 1. IS732 Covers the essential requirements and precautions regarding wiring installations for ensuring satisfactory and reliable service and safety from all possible hazards from the use of electricity 2. Applicable to design, selection, erection, inspection and testing of wiring installations whether temporary or permanent 3. Relates to all wiring installations in non-industrial and industrial locations (ii) 1. Contains guidelines on choosing proper size of various components of earthing system especially earthing conductors (SWG) and earthing 2. Gives guidance on the methods which are adopted to earth and electrical system for limiting the potential of current carrying conductors forming part of the system ie. System earthing and Equipment earthing 3. This code applies to land based installation and it does not apply to ships, air crafts or offshore installations	(8)
		OR	
12	a)	The Indian Electricity Act, 2003 seeks to replace Indian Electricity Act 1910, The Electricity Supply Act 1948, and The Electricity Regulatory Commission Act 1998 It is an act implemented by the parliament to consolidate the laws relating to generation, transmission, distribution, trading and use of electricity and aims at • Promoting measures to the development of electricity industry. • Promoting competition • Protecting interest of consumers • Providing electric supply to all areas • Providing transparent policies on subsidies • Constitution of Central Regulatory Authority and Regulatory Commission • For all matters connected to therewith and incidental there to The salient features of the act are: • Delicensing of generation • Liberalization of captive power policy	(8)

		• Open access to transmission and distribution network • Stringent penalties for power theft • Transparent subsidy management • Thrust on Rural electrification	
	b)	Replacing existing laws while preserving the core features Introducing new concepts like power trading, open access To prevent the requirement of each SEB's to pass their own act Give SEBs to develop their own power sector Include progressive features and endeavours	(6)
		Module II	
13	a)	The ratio of lumens reaching the working plane to the total lumens given out by the lamps is known as CoU or Utilization factor CU depends on the following factors – Lumen output of the fitting – Shape of the room – Reflection factors of walls and ceilings – Height of the ceiling – Arrangement of the fittings etc	(8)
	b)	$I = Eh^2 = 254$ Cd $E = \frac{I \cos\theta}{d^2}$ =32.5 lux	(6)
		OR	
14	a)	**Advantages of LED Lamps for Household Applications:** **Energy Efficiency:** LEDs are highly energy-efficient, converting a higher percentage of electrical energy into light. **Long Lifespan:** LED lamps have a longer lifespan compared to traditional incandescent and fluorescent bulbs.. **Durability:** LEDs are solid-state lighting devices, making them more durable and resistant to breakage. **Instant Light:** LEDs provide instant full brightness without the warm-up time required by some other types of lighting. **Cooler Operation:** LEDs emit very little heat in comparison to incandescent bulbs, which release a significant amount of energy as heat. **Disadvantages of LED Lamps for Household Applications:** **Initial Cost:** LED lamps can have a higher upfront cost compared to traditional bulbs, although the prices have been decreasing over time. **Sensitivity to Temperature**: Extreme temperatures can affect the performance of LED lamps. **Color Rendering:** While advancements have been made, some people still prefer the color rendering of traditional incandescent bulbs.	(6)

b) $Hm = 8 - (1.5 + 1) = 5.5m$ (8)

$SHR = 1.5 : 1$

Therefore, $S = 1.5 \times 5.5 = 8.25m$



Min. no. of rows $= \frac{W}{S} = \frac{20}{8.25} = 2.4$ (3 rows)

Min. no. of luminaires per row $= \frac{L}{S} = \frac{40}{8.25} = 4.85$ (5 luminaires)

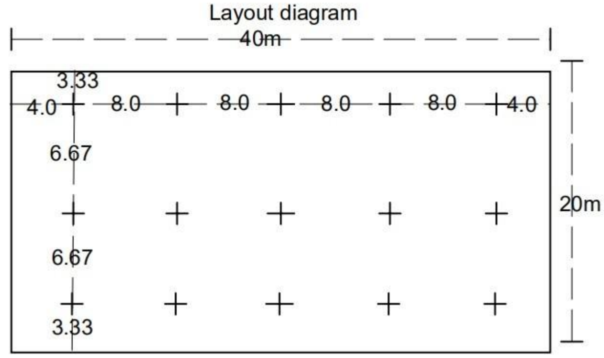
This means that the minimum number to conform with SHR. requirement is 3 rows with 5 luminaires per row.

Assuming that three rows of five luminaires is suitable, the actual spacing is determined as follows:

Spacing between rows (S) $= \frac{W}{\text{no.of rows}} = \frac{20}{3} = 6.67m.$

Spacing in rows (S) $= \frac{L}{\text{No. per.row}} = \frac{40}{5} = 8m$

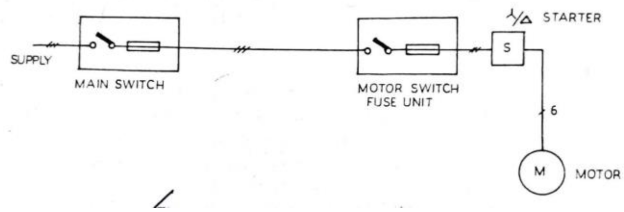
Layout diagram

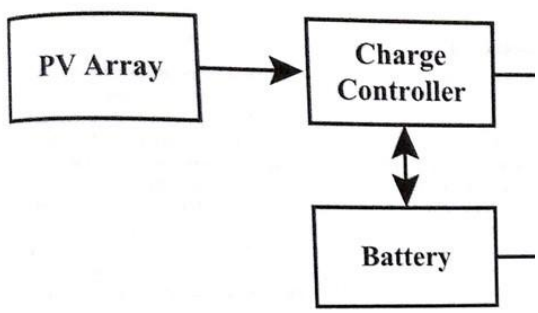
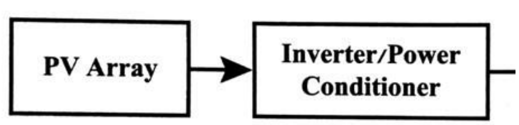


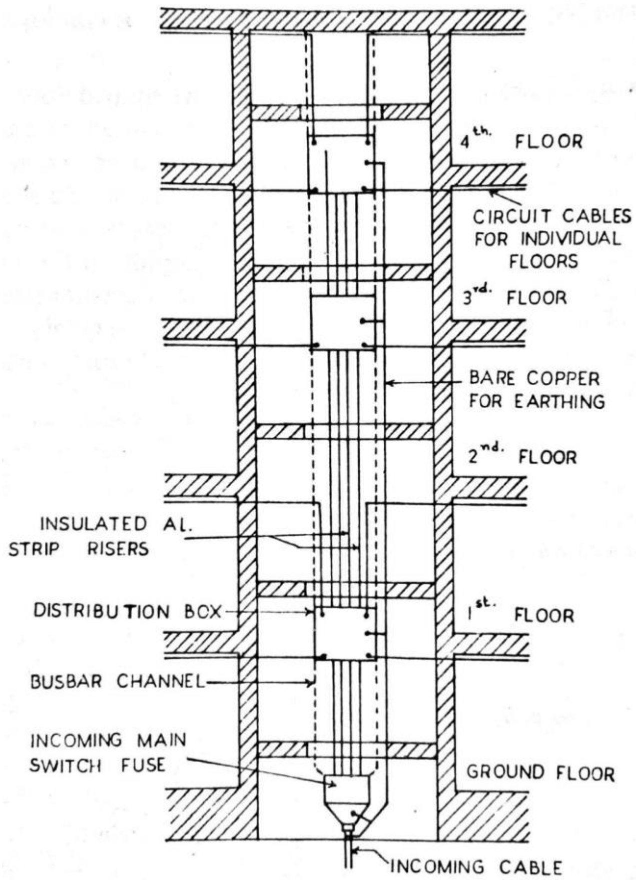
		Module III	
15	a)	RCDs detect an imbalance in the live and neutral currents. A current overload, however large, cannot be detected. •i.e. RCDs don't offer protection against current overloads It is now possible to get an MCB and RCD in a single unit, called an RCBO •RCBOs are commonly used in applications where there is the need to combine protection against over current (overload and short-circuit) and protection against earth leakage currents. •They can generally be fitted into a Consumer Unit in place of an MCB. •Some are the same width, others are wider and may take up two spaces, this depends on the brand and model •They tend to be quite expensive. •However, in the respect that, in the event of an earth leakage, they will cut power only to the affected circuit without inconveniencing users on other circuits •They are superior to a split-load arrangement or an installation protected by a single RCD, because either of these will cut power to the other circuits	(4)
	b)	Total Connected Load = Light Load + Power Load = 4633 W Since total Connected Load is less than 5kW single Phase supply can be Provided. Light sub circuit Number of light sub circuits = No of light points/8 = 25/8 = 3.1(4 circuits) Power sub circuit Total number of power sub circuits = 1 for 16A PP+1 for water Heater+1 for Pump = 3 Total number of sub circuits = light circuits + Power Sub circuits = 4+3 = 7 Select 230V,single phase 8 way DB(standard DB)	(10)

		Incomer: 25A RCBO or RCCB with 30mA sensitivity Outgoing MCBs: 4 number of 6A MCBs for 4 light sub circuits 3 number of 16A MCBs for 3 Power Sub circuits MCBs are selected based on maximum current flowing in sub circuit Maximum current in a light sub circuit, $i = p/ v\cos\theta = 3.47A$ Maximum current in a Power sub circuit, $i=p/v\cos\theta =13.04A$ Hence it is a thumb rule to use 6A MCB for light sub circuit and 16A MCB for Power sub circuit. In case of motor load, use MCB with Ampere rating greater than starting current. selection of Main switch Fuse Unit (SFU) Fuse is selected based on total load current $i=P/V\cos\theta = 4633/230x1 = 20.1A$ 25A fuse is selected. SFU rating must be greater than fuse rating. Hence 32A,Iron Clad Double Pole SFU with 25A fuse can be used. For light sub circuits : 1.5mm sqr Copper wire For Power sub circuits : 2.5mm sqr Copper wire For earth wiring : 1.5 mm sqr Copper wire + Line diagram of the arrangement	
		OR	
16	a)	The feeder to a DB will be fed from SDB or MDB. It is necessary that the major protective fuse shall not blow off before or along with motor protective fuse in the DB. There shall be grading or discrimination between these fuses. The fuses in two systems are termed as minor and major fuses. For achieving grading, the minor fuse should be lower than half of the major fuse. Eg: (1) 32A will grade with 63A and above	(4)
	b)	Step 1- Draw a suitable plan as per the details given above. Step 2- Place the lights, fans, sockets etc as per IS standard	(10)

		Step 3 - Calculate the total connected load and decide the number of power and light sub-circuits. Step 4 - Draw the layout and wiring diagram Step 5 - Decide the Switches, Breakers, ELCB, and DB for the above. Step 6 - Calculate the length of phase wire, neutral wire and earth wire Step 7 - Decide the size of phase wire, neutral wire and earth wire Step 8 - Estimate the quantity of materials required.	
		Module IV	
17	a)	1. Insulation resistance test (IR test) 2. Dielectric absorption and polarization index test: 3. Two voltage test 4. Measurement of tan delta 5. Transformer ratio test 6. Measurement of short circuit current. 7. Measurement of magnetizing current 8. Test of transformer oil 11. Temperature Indicator Test 12. DC Winding Resistance Test	(4)
	b)	As it is densely populated area, indoor substation is preferred. An 11kV/415V, 300kVA indoor type transformer with HT and LT cables can be used. To find the size of cable to be run from the secondary side of the transformer, the current rating should be found out. Full load current = $I_L = \frac{300 \times 10^3}{\sqrt{3} \times 415} = 417A$ Taking a factor of safety, 2 runs of 3-1/2 core 400sq.mm PVC /XLPE insulated LT cable can be used. The LT panel is having: One incoming line of 500kVA OCB Single line diagram: Indoor substation Estimate of the materials:	(10)
		OR	
18	a)	The current rating of cables for supply to motor may be based upon the normal full load current of the motor, but the rating of fuse should be based upon the starting current and type of strting	(4)

			
	b)	Main supply: Assume the motor efficiency to be 85% and pf to be 0.8 for the machines. $$\text{Motor 1, } I_{L2} = \frac{hp \times 746}{\sqrt{3} \times V \times \eta \times pf} = \frac{3 \times 746}{\sqrt{3} \times 415 \times 0.85 \times 0.8} = 4.5A$$ $$\text{Motor 2, } I_{L3} = \frac{hp \times 746}{V \times \eta \times pf} = \frac{0.5 \times 746}{240 \times 0.85 \times 0.8} = 2.25A \text{ --- 1 mark}$$ $$\text{Motor 3, } I_{L4} = \frac{hp \times 746}{\sqrt{3} \times V \times \eta \times pf} = \frac{1 \times 746}{\sqrt{3} \times 415 \times 0.85 \times 0.8} = 1.5A$$ Motor 1, I_{L2}= 4.5A, I_{s2}= 6.75A, 3/0.737mm Cu cable (10A) Motor 2, I_{L3}= 2.25A, I_{s3}= 3.37A, 1/1.80mm AL cable (10A) Motor 3, I_{L4}= 1.5A, I_{s1}= 2.25A, 3/0.737mm Cu cable (10A) Size of conduit for each motor Rating of switch fuse : The 2 nos. 3 phase motor of rating 3,1hp can be controlled by 16A, 500V TPIC switches and the 1/2 hp single phase motor can be controlled by a 16A, 250V DPIC SFU.	(10)
		Module V	
19	a)	1. Minimum 1m clearance shall be provided on three sides of the generator set, minimum 2m clearance shall be provided between them 2. Fuel tank of DG sets shall be installed outside the generator room 3. Exhaust pipe of DG sets shall maintain a minimum height of 1.8m clearance from the floor level 4. Voltmeters and frequency meters shall be connected before the circuit breaker in the generator control panel 5. Watt hour meter and ammeters in each phase shall be provided 6. For generators of 500kVA and above, kVA/kW meter and pf meter shall also	(6)

		be provided 7. Changeover switches of approved makes is permitted for capacities up to and including 800A 8. Above 800A capacity, CB with castle key interlock shall be provided 9. Under voltage coil may be provided at grid side, ACB/MCCB on respective switchboards at installations where generators are installed for preventing chances of back feeding from generators to grid side 10. Separate Isolators shall be provided before the change over switch, if it has no OFF position 11. Provide thermal O/L relay in the generator circuit 12. Generator capacity should be sufficient if it is called upon to start induction motors of 60% of its rating, provided suitable starters are employed for limiting the starting current. 13. Generator room shall be made of non-inflammable materials 14. Standard size of earthing cables and conductors for generators shall be provided 15. Proper protection schemes should also be provided for generators 16. Central pollution control board approved type acoustic enclosure shall be provided for all generators	
	b)	On-grid system  Off – grid system  	(8)
		OR	

20	a)	 	(4)
	b)	3 unit energy = 3 kWh consumption Watt-hr per day to be generated by the solar panels = 1.3 x 3000 = 3900 W-hr Let the size of the solar panel = 4.83 kWh/m^2/day Panel generation factor (with MPPT controller) = 4.83 x derating factor = 4.83 x 0.69 = 3.33 Wh/Wp/day Watt hr/day for all the appliances taken together = 3000 Wh/day Panel size required = 3900/3.33 = 1171.17 Wp Therefore, number of PV modules = Panel size required / Rated Wp of the PV module Choose 110Wp module Number of PV modules = 1171.17 / 110 = 10.65 (say 10 modules) The capacity of the solar plant is therefore 10 x 110 = 1100 Wp The surface area of each module is 0.85 m^2	(10)

Therefore the roof area required for installing the solar PV system will be about $10 \times 0.85 = 8.5m^2$

Size of the battery

Wh/day required by the load = 3000

Battery must supply 3000 / 0.85 = 3529 Wh/day (based on 85% efficiency)

Battery must supply 3529/ 0.5 = 7058 Wh (based on 50% depth of discharge)

Battery must be sized 7058 x 2 = 14117 Wh/day (based on 2 cloudy days)

Say 14100 Wh.

With 24V battery, Ah capacity = 14100 / 24 = 587.5 Ah (say 600Ah)

Therefore we can choose 8 nos. Of 150Ah, 12V batteries connected as two in series and four in parallel (4 x 150 = 600 Ah at 24 V) and this will be able to supply the load for 2 days without interruption.

Size of charge controller

Since the battery voltage is selected as 24V, the PV modules shall be connected as two numbers in series and five numbers in parallel. so that the open circuit voltage of the PV module will be around 21 x 2 = 42 V and the short circuit current of each module is 6.98 A (from the table)

Therefore the total short circuit current from all the 5 PV modules connected in parallel will be equal to = 6.98 x 5 = 34.9 A

Hence the current rating of the charge controller = 130% of the short circuit current

= 1.3 x 34.9 = 45.37A (say 50A)

Therefore we can choose a charge controller rated for 24V, 50A capacity

The power output of the PV array is 1100W. hence the capacity of the inverter can be fixed as 1200VA

The final specification of the inverter can be 24V DC inputs, 240V AC output, 1200VA grid connected inverter.